Mohamed Abrar. S. K.

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Professional Summary

Final-year B.Tech student in Electronics and Communication Engineering with practical experience in Embedded Systems, IoT prototyping and Digital Design. Completed an internship in Digital Design and Verification using Verilog HDL. Passionate about developing embedded hardware, industrial automation solutions and real-world engineering prototypes while continuously expanding technical knowledge.

Education

Kalasalingam Academy of Research and Education Expected Graduation: 2027
Bachelor of Technology (Electronics and Communication Engineering) CGPA: **8.06* / 10**
Minor: Aeronautical Engineering (Unmanned Aerial Vehicles)

Technical Skills

Programming	Embedded-C, C++(basics), Python(basics).
Core Skills	Digital Circuit Design, Peripheral Connectivity and Configuration, Embedded Systems, Industrial Robotics, Digital Electronics, Power Electronics, Electric Vehicles
CAD Tools	OnShape, TinkerCAD,

Internships

MSPAND Technologies Pvt. Ltd. May 2025 – June 2025
Digital Design and Verification Intern

- Completed internship in Digital Design and Verification using Verilog HDL.
- Designed and simulated a Traffic Light Controller using Verilog HDL.
- Implemented and verified a 64-bit Arithmetic Logic Unit (ALU).
- Gained practical exposure to RTL design, simulation, debugging and hardware verification.

Projects

Programmable Universal EV Controller (Ongoing) Dec 2025 – Present
Technologies: Embedded Systems, Power Electronics

- Designing a programmable universal EV controller compatible with multiple electric two- and three-wheelers.
- Developing a configurable power and control architecture for different motor and battery configurations.
- Working on embedded hardware development, controller prototyping and power electronics integration.

Cloud Controlled Robotic Arm using STM32F103RB Jul 2025 – Nov 2025
Technologies: STM32, Embedded Systems

- Developed a cloud-controlled robotic arm with real-time video monitoring.
- Responsible for STM32 microcontroller interfacing and embedded hardware implementation.
- Integrated peripheral devices and validated the complete prototype.

Robotic Arm using STM32F103RB Jan 2025 – Apr 2025
Technologies: STM32, Embedded Systems

- Led the hardware design and STM32 peripheral interfacing.
- Developed embedded firmware for actuator control.
- Formed the foundation for the cloud-controlled robotic arm.

IoT Attendance Monitoring System using ESP32 T-Display S3 Jul 2024 – Nov 2024
Technologies: ESP32, IoT

- Developed an IoT-based attendance monitoring system using the ESP32 T-Display S3.
- Implemented hardware integration and peripheral interfacing.
- Tested and validated the complete embedded prototype.

Electricity Generation using Piezoelectric Crystals Dec 2023 – Apr 2024
Technologies: Power Electronics

- Developed a prototype demonstrating electricity generation using piezoelectric crystals.
- Led hardware development and coordinated testing activities.
- Gained practical experience in energy harvesting and basic electronics.

3D Printed Mobile Charger with Holder Jul 2023 – Nov 2023
Technologies: OnShape, CAD

- Designed a 3D printed mobile charger holder using OnShape software.
- Applied enclosure design principles and component fitting techniques.
- Improved practical knowledge of mechanical product prototyping.

Patents

PyroSentinal Feb 2026

- AI-powered LoRa-based low-power forest fire early detection system.
- Designed an intelligent communication framework using Particle Swarm Optimization (PSO)-optimized mesh architecture.
- Patent Filed.

AI-Based Interactive Smith Chart May 2026

- Developed an AI-based interactive Smith Chart for simplified digital learning and visualization.
- Improves understanding of Smith Chart concepts through an interactive learning approach.
- Patent Filed.

Hackathons

IoT Crowd Monitoring using N8N FlowFusion Feb 2026
Achievement: First Prize

- Developed an IoT-based crowd monitoring system integrated with automated mobile notifications.
- Implemented workflow automation using N8N FlowFusion.

Intelligent Toll Booth using Cadence Virtuoso Nov 2025
Achievement: Third Prize

- Designed and developed an intelligent toll booth prototype using Cadence Virtuoso.
- Gained hands-on experience in analog and digital circuit design.

Blood Sugar Monitoring System Nov 2025

- Developed a GUI-based health monitoring system for blood sugar visualization.
- Designed the prototype for future wearable smart health monitoring applications.

Car Dashboard System Sep 2024

- Designed a real-time dashboard displaying speed, RPM, engine temperature and tyre pressure.
- Developed an interactive graphical interface for monitoring vehicle parameters.

Professional Memberships

- ACM India Member
- KARE ACM SIGBED Student Chapter Member

Soft Skills

Analytical Thinking	Team Leadership
Problem Solving	Adaptability
Self-Learning	Communication

Areas of Interest

Embedded Systems • Industrial Automation • Internet of Things (IoT) • Electric Vehicles • Firmware Development